Synthese (2011) 183:389–408
DOI 10.1007/s11229-010-9869-1

Mechanistic explanation at the limit

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Received: 27 August 2010 / Accepted: 31 December 2010 / Published online: 19 January 2011

Abstract Resurgent interest in both mechanistic and counterfactual theories of explanation has led to a fair amount of discussion regarding the relative merits of these two approaches. James Woodward is currently the pre-eminent counterfactual theorist, and he criticizes the mechanists on the following grounds: Unless mechanists about explanation invoke counterfactuals, they cannot make sense of claims about causal interactions between mechanism parts or of causal explanations put forward absent knowledge of productive mechanisms. He claims that these shortfalls can be offset if mechanists will just borrow key tenets of his counterfactual theory of causal claims. What mechanists must bear in mind, however, is that by pursuing this course they risk both the assimilation of the mechanistic theories of explanation into Woodward's own favored counterfactual theory, and they risk the marginalization of mechanistic explanations to a proper subset of all explanations. An outcome more favorable to mechanists might be had by pursuing an *actualist-mechanist* theory of the contents of causal claims. While it may not seem obvious at first blush that such an approach is workable, even in principle, recent empirical research into causal perception, causal belief, and mechanical reasoning provides some grounds for optimism.

Keywords Explanation · Mechanisms · Counterfactuals · Causation · Causal perception · Mechanical reasoning · Simulation

1 Introduction

Resurgent interest in both mechanistic and counterfactual theories of explanation has led to a fair amount of discussion regarding the relative merits of these two approaches

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(Bogen 2004; Glennan 2002, 2010; Machamer 2004; Psillos 2004; Woodward 2002, 2003, 2004). At present, James Woodward appears to be the pre-eminent counterfactual theorist, and he has put mechanists on the defensive regarding (among others) two important issues. He claims, to a first approximation, that mechanists have not been able to make sense either of explanations put forward absent knowledge of mechanisms (viz., stand-alone causal claims) or of claims about causal interactions between mechanism parts (i.e., in cases where explanations do cite specific mechanisms). Woodward suggests that mechanists can remedy these defects by borrowing key tenets of his counterfactual theory. This may be so, but, I argue, the price of accepting this offer may be the assimilation (into Woodward's counterfactual framework) and marginalization (to a proper subset of explanations) of the mechanistic approach to explanation.

Here I consider another potential remedy for what ails mechanists—namely, an *actualist-mechanist* theory of the contents of causal claims. Woodward has already criticized this option, but only under the assumption that actualist-mechanist theories require that causal claims make assertions about specific productive mechanisms. In fact, an actualist-mechanist may instead propose that although causal claims always assert something about actual productive mechanisms, they in fact never (or hardly ever) assert anything about *specific* mechanisms. Still, at first glance it is hard to see how this proposal could be fleshed out into something remotely respectable. Searching for a way to do just that, I draw upon multiple lines of recent research in experimental psychology.

Like Woodward (forthcoming), I assume that there is an intimate connection between our concept of causation (and the beliefs, suspicions, etc. in which it figures) and the contents of our causal claims. I argue that a body of research on topics ranging from infant causal perception to adult mechanical reasoning suggests that our concept of causation may (at least initially) just be an abstraction from instances in which we integrate a variety of early perceptual expectations in order to understand how events in specific mechanical systems unfold. This account allows for, without giving pride of place to, an ability to determine the consequences of counterfactual interventions. Woodward has, however, also expressed a number of concerns regarding actualist-mechanist theories of our *concept* of causation, including the version offered here, and so I attempt to address some of what I take to be the most pressing.

Having offered a fleshed-out version of the non-specific, actualist-mechanist theory of our concept of causation, I propose that it lends itself to a corresponding theory of the contents of causal claims, with all of the accompanying benefits. I focus, in closing, on how a theory of this sort might answer the two major criticisms of mechanistic theories of explanation mentioned earlier and, thereby, stave off Woodward's apparent attempt at assimilation and marginalization. None of this, admittedly, is meant to provide a decisive refutation of Woodward's criticisms of mechanistic theories of explanation. Rather, my main intent is just to show that there is at least one way, a way that seems to have gone unexplored, in which mechanists might respond.

Before getting under way, it is worth noting that there are at least three prominent views regarding what sorts of things explanations are that have been adopted by latter-day mechanists, each arguably reflecting a common manner of speaking (Waskan 2006). Some adopt Salmon (1998) *ontic* conception and equate explanations with the objective mechanisms that produce the happenings that perplex us (Glennan 2002;

Craver 2007). Others take explanations to be cognitive states constituted in part by representations of possible productive mechanisms (Nersessian 2002; Waskan 2006, 2010; Wright and Bechtel 2007; Horst 2007; Thagard 2010). Still others equate explanations with the representations, often termed 'models,' that are used to communicate and disseminate information about possible productive mechanisms (Machamer et al. 2000; Craver 2006). The criticisms of the mechanistic approach with which I will be grappling here are directed, at least in large part, at theories of this last, communicative sort.

2 Woodward's criticisms

Getting down to business, two of Woodward's biggest concerns about the mechanistic approach to understanding the contents of explanations (by which is hereafter meant sets of explanatory claims) are the following:

(i) All explanations that do make claims about mechanisms bottom out at causal claims that are best understood in terms of counterfactuals (Woodward 2002).
(ii) Some explanations are not constituted by claims about mechanisms (Woodward 2004).

The first worry concerns what might be called *mechanism explanations*, which are sets of claims that explicitly describe the mechanisms that might be responsible for producing the happenings that perplex us. Consider, for example, the following explanation, offered by Mink (1996), for movement disorders exhibited in late-stage Huntington's patients (viz., progressive difficulty in the execution of movements) and in a certain class of Parkinson's patients (viz., co-contraction rigidity). Both of these are associated with pathologies of the basal ganglia (a set of nuclei lying deep within the brain). The first involves degeneration of neurons of the striatum and the second involves loss of inhibitory neurons of the *substantia nigra*.

To explain the relationship between these neuropathologies and movement disorders, Mink puts forward a set of descriptions and depictions (a model) of the wiring of the relevant cortical and sub-cortical structures. To vastly oversimplify, he claims that motor-planning systems of the cortex initiate movement by focally exciting neurons in the striatum and globally exciting neurons of the *substantia nigra*. The latter in turn send inhibitory signals to each of the endogenously active neural triggers (think of a switchboard) in the cerebellum for familiar motor patterns (e.g., raising one's arm, clenching one's fist, etc.). At the same time, local activity in the striatum leads to selective inhibition of the *substantia nigra*, which has the effect of selectively *dis*inhibiting specific motor-pattern triggers. It is easy to envision how, on this setup, degeneration of striatum might produce movement-execution deficits (i.e., no brake would ever be disabled) and how degeneration of SN would lead to co-contraction rigidity (i.e., many brakes would be disabled simultaneously).

Mechanists are somewhat divided on the details (see Tabery 2004), but they tend to construe such mechanism explanations as sets of claims to the effect that the (regular or singular) happenings that perplex us are the product of *parts* (or components) that are arranged, act (or change), and *interact* with (or effect changes in) one another. According to Woodward (2002), problems arise for mechanists insofar as they attempt to make sense of what the claims about part *interactions* come to without invoking counterfactuals. Some of these claims take the form of what I will call *bare* causal claims (i.e., claims of the form *x causes y*), and others have, at least superficially, a more *contentful* form (e.g., *x inhibits y, x oxidizes y*, *x flips y,* etc.) (see Bechtel and Richardson 1993; Machamer et al. 2000; Machamer 2004). Woodward rightly notes that mechanists owe us an account of the contents of these claims, of what they assert about the world in virtue of which they contribute to mechanism explanations.

The alternative Woodward offers up is his "interventionist account," which, "connects causal claims to counterfactual claims about what *would happen* if (perhaps contrary to actual fact) certain interventions *were* carried out" (2004, p. 59). His proposal appears to be roughly that a type-causation claim to the effect that X causes Y asserts that the values or magnitudes of X and Y are such that there are possible conditions under which, when all else is held constant, intervening on X (which may not be humanly possible) would (via X, rather than some alternate route from intervention to Y) alter Y (or its at least its probability distribution) (Woodward 2002). For instance, to claim that raising/lowering atmospheric pressure (A) causes an increase/decrease in the readings of an appropriately situated mercury barometer (B) is largely to assert that under various interventions on the former the correlation between it and the latter would continue to obtain. Of course, as with any high-level regularity, there will be countless conditions under which the correlation will fail to obtain (e.g., the pressure increases to an astronomical level, the temperature falls to $-39\,$C, the instrument is smashed with a hammer, etc.). Accordingly, when we make a high-level causal claim such this one, it is understood or implied that we are not asserting that the correlation is completely invariant, just that it is "approximately" invariant (Woodward 2002, pp. S368–S370). Thus, one might say that the causal claim, "A causes B," asserts that an *Approximate Counterfactual-Intervention Dependence* (ApCoInD) relation holds between the values of A and B. As for token-causation claims, to say that an instance of A (a) caused an instance of B (b) is, I gather, mainly to assert that there are some possible (but non-actual) interventions on a that would have (via a) altered b (Woodward 2004, p. 46).

Woodward claims that there are several benefits to this way of making sense of bottom-out claims regarding part interactions (YR). He claims, for instance, that it weeds out illegitimate bottom-out claims (i.e., those that hold only accidentally or as a result of a common-cause) and that it accommodates the narrow scope and exceptionful character of many legitimate bottom-out claims. A number of prominent thinkers appear to have been at least partly swayed by such considerations (see Glennan 2002; Psillos 2004; Craver 2007).[1] Indeed, it has to be admitted that Woodward has offered up a promising cure for what ails mechanists. Mechanists should bear in mind, however, that on Woodward's view mechanism explanations are explanatory precisely in virtue of making fine-grained assertions about where the happenings that perplex us fit within a nexus of alternative possibilities. They render happenings intelligible by "enabling us to see how, if … conditions had been different or had changed in various ways, various of these alternative possibilities would have been realized instead" (Woodward 2003,

[1] Lately, Glennan (2010) has seemed to qualify his earlier endorsement of Woodward's position.

p. 191). Put differently, a "model in effect encodes a set of predictions about what will happen in various hypothetical experiments…" (2002, p. 376). In short, it appears that on Woodward's view mechanism explanations just reduce to networks of claims asserting ApCoInD relations (2002, p. 375; 2003, pp. 222–223; 2004, pp. 65–66).

Perhaps Woodward is right about all of this, but at the very least it seems to run against the grain of what most mechanists believe, or at least once believed, which is that a mechanism explanation for some happening that perplexes us is explanatory precisely in virtue of its capacity to enable us to understand how the parts of some system actually conspire to produce that happening. I doubt that many mechanists would deny that we can use mechanism explanations, or the understanding they provide, to make inferences about counterfactual conditions, but I should have thought that this is considered of ancillary importance, having to do with how we evaluate our explanations rather than with what makes them explanations in the first place.

Woodward's second criticism is that explanations that are not constituted by claims about the underlying or intervening mechanisms of action. There are, in particular, explanations constituted primarily by claims about causal interactions (e.g., "Taking mandrake root cures infertility") of whose underlying mechanisms of action we are entirely ignorant. Thus, says Woodward, "Fans of mechanisms owe us an account of what it is that we've established when we learn, e.g. that a drug cures some illness but don't yet know about the mechanism by which it does so. Counterfactual accounts provide a straightforward answer to this question" (2004, p. 66). This criticism has the potential to be far more damaging than the first. Whereas the first may lead to what some mechanists will view as a healthy amendment to their theory, the second more clearly relegates the mechanistic approach to the inferior status of handling only a proper subset of explanations.

Taking stock, then, Woodward contends that insofar as mechanism explanations are concerned, the mechanistic approach needs, and ought ultimately to be assimilated into, the counterfactual approach. He would have it, moreover, that the mechanistic approach is relevant to only a portion of the total explanation pie. The counterfactual theory, in contrast, is said to provide a unified account of all explanations. In sum, what Woodward seems to have in mind is the assimilation and marginalization of the mechanistic approach to explanation. Before conceding so much, let us consider whether or not there are any alternative strategies for dealing with these concerns that paint the mechanistic approach in a more positive light.

3 Alternative strategies

Notice, to start with, that the causal claims at issue in (ii) are the same basic sorts of claims at which mechanism explanations (at least often) bottom out. Thus, a viable mechanist response to (ii) will in all likelihood answer (i) as well. Notice also that just about any theory of causation will help mechanists at least stem the bleeding with regard to (i), and this may be a large part of the reason why mechanists have been receptive to Woodward's proposition. However, if they had a viable non-counterfactual (or *actualist*) theory of causation, mechanists about explanation could decisively stave off Woodward's apparent attempt at assimilating their approach into his. Moreover, in order to answer (ii) and, thereby, resist Woodward's attempt to marginalize the mechanistic approach to explanation, it would seem that not only is an actualist account of the contents of causal claims needed, but an actualist-*mechanist* account is needed.[2]

3.1 Alternative 1: a specific actualist-mechanist theory of causal claims

The driving intuition behind some mechanists' resistance to Woodward's account of the contents of causal claims is that causal claims make assertions about what *actually* happens rather than about what would happen (Bogen 2004; Machamer 2004; also see Wolff 2007). In particular, some contend that causal claims make assertions about actual productive *mechanisms* connecting cause and effect. One thus might, in response to (ii), claim that bare and contentful causal claims are explanatory in virtue of *this* fact. Moreover, if this account of the contents of these causal claims proved viable, it would also provide mechanists with a ready response to (i).

To see, more specifically, what such an account might look like and where it might get into trouble, consider the following causal claim:

(1) Degeneration of dopaminergic striatal neurons (S) causes movement-execution difficulties (E).

There are at least two distinct ways in which we might conclude that (1) is the case. On the one hand, we might monitor the relative values of S and E and come to believe on this basis, and finally to claim, that S causes E. Let us call this a *superficial* justification. For our purposes, it matters not whether our observations are passive (e.g., nature sets the values of S) or active (e.g., humans intervene on the values), so long as the general pattern is that when S occurs so does E and (let us assume) that when E fails to occur so does S. Woodward seems to agree that such observations would lend support to (1). He would, I gather, propose that under these evidentiary conditions the causal claim roughly asserts that it is approximately true that were S increased or decreased this would, respectively, (via S) increase or decrease E. As will become clearer by way of contrast, one might say that causal claim under these conditions asserts what might be termed a *coarse-grained* ApCoInD relation (see Woodward 2004).

One might, in principle, also reach the conclusion that (1) is true by learning about the relevant physical parts and processes of the nervous system (see Glennan 1996, p. 66). For instance, one might discover that Mink's general proposals about the mechanisms connecting motor planning centers and motor pattern triggers are correct and subsequently determine that an implication of this setup is that increasing or decreasing S will result—that is, absent such impediments as the introduction of reuptake inhibitors, electrocortical stimulation, etc.—in an increase or decrease in E.

One way mechanists might try to at least gain a foothold on the issue of what causal claims assert would be to suggest that, at least when (1) has this sort of *deep*

[2] I am here granting the central premise of (ii), which is that some genuine explanations assert little more than that the happening in question is the effect of some cause. One might contest this point, but here I will try to show that all may not be lost for mechanists even granting this assumption.

justification, it makes an assertion about the specific mechanisms by which changes in S bring about changes in E. On this sort of approach, which Woodward seems to anticipate, "once intervening mechanisms are filled in… an appeal to counterfactuals is no longer necessary" (2004, p. 65).

This appears to be very much along the lines of what was originally proposed by Glennan (1996), who seemed to think that the high-level regularities that characterize causal interactions between mechanism parts are just the public face of the underlying mechanisms that sustain *them*. Thus, the rough idea is that S causes E if and only if the two are "connected in the appropriate ways" by mechanisms (p. 56). This is an ontological thesis, but Glennan (1996, 2010), wishes to see it carried over into the semantic realm as well, and thus he might say that (1) *asserts that* S and E are connected in appropriate ways by mechanisms. More specifically, what (1) asserts on this view is that S produces E through a continuous chain of interactions between the parts of some system. On one natural interpretation of this proposal, what one knows of the underlying mechanisms would, under these deep justificatory conditions, inform, and *affect the content of*, one's claim that the one type of happening causes the other.

Woodward's main reaction to this proposal is to point out that the actualist-mechanist still has trouble accounting for the fact that we sometimes establish that a causal relation holds even without "provision of extensive information about mechanisms" (2004, p. 66). In other words, even if this actualist-mechanist proposal could make sense of causal claims made under deep justificatory conditions, it still has trouble with shallow conditions. Woodward, on the other hand, can claim to have offered a single, counterfactual account of what both superficially and deeply justified causal claims assert.

I mention all of this in part to make clear that Woodward's worries about the possibility of an actualist-mechanist approach to causal claims are, perhaps for good exegetical reasons, based upon the assumption of what might be called a *specific* reading of the approach, a reading on which causal claims always make assertions about specific mechanisms of action. On this reading, (1) might be taken to always assert that the occurrence of S in some specific mechanical system produces, through a continuous chain of part interactions, E. The worry, however, is that in those cases where we possess no information about the specific mechanisms of action, *there is nothing for the putative mechanical assertion to assert*. If there's a way past this concern about actualist-mechanist theories, it may well begin with the rejection of the idea that causal claims always make assertions about *specific* mechanisms.

3.2 Alternative 2: a non-specific actualist-mechanist theory of causal claims

To see how this alternative might go, notice first that Woodward himself has a couple of options regarding how to interpret (1). On the one hand, he might opt for a semantic-continuum approach whereby varying the depth of the justificatory information alters what (1) asserts about the world—in particular, the deeper the evidentiary information, the finer the network of ApCoInD relations asserted. Although Woodward could say, in accordance with this approach, that (1) asserts ApCoInD relations under both superficial and deep evidentiary conditions, the approach would not entirely univocal in that the precise content of the relevant assertion will still vary with the depth of evidentiary information.

On the other hand, he might adopt a genuinely univocal approach according to which (1) always assert the very same thing regardless of the depth of evidentiary information. Of course, since we do not always possess information about fine-grained ApCoInD relations, it would make little sense to say that what a claim such as (1) *always asserts* is that certain fine-grained ApCoInD relationships obtain. Another (perfectly live) option would be a full-shallow version of the approach on which (1) *always* make a very non-specific assertion to the effect that E bears an ApCoInD relation to S. Let us call this a *non-specific* ApCoInD theory of causal claims.

Actualist-mechanists have similar options available to them. They might propose that the deeper the information we have to back a claim like (1), the more specific the assertion about productive mechanisms. Then again, they might go for a more univocal approach whereby what (1) asserts is the same regardless of depth of information. As with the non-specific ApCoInD theory, the challenge here is still to formulate a version that is compatible with purely shallow evidentiary information. The problem, of course, is that we still lack an actualist-mechanists account of what causal claims assert under the shallowest evidentiary conditions. Indeed, if we had one, we could use it to anchor the shallow-most endpoint of the aforementioned semantic-continuum. Bearing this in mind, I will nevertheless concentrate my efforts on determining whether or not there is a viable, or potentially viable, *consistently* non-specific actualist-mechanist—I will use the partial acronym *AcMe*, under the assumption that it also connotes a lack of specificity—theory of causal claims, one that is analogous to the non-specific ApCoInD theory.[3]

To be compatible with purely shallow evidentiary (or other) information, the idea behind the AcMe theory would have to be that (1) always (or at least nearly so) makes a non-specific assertion to the effect that S bears *some general sort of mechanical relation* to E. The tricky part, of course, is figuring out how to flesh this idea out into something useful. I will not take a decisive stand here on precisely how to accomplish this, but I will consider one possibility that I find highly promising. The immediate point of the exercise will be to show that, despite how it may appear at first blush, there may be a way to put flesh on the skeletal notion of an AcMe theory. The proposal considered here originates with research into the psychology of causal perception and belief, to which I now turn.

[3] It might be argued, moreover, that the right course is to pursue a univocal approach, regardless of what general sort of theory of causal claims one favors. *Prima facie* what a causal claim asserts to be the case in a particular situation will not always track everything the speaker *believes* to be the case about the detailed underlying or intervening mechanisms. For instance, it would be odd to suggest that (1) ever asserts anything about the wiring of the basal ganglia, even if the speaker is someone like Mink who knows a great deal about the mechanisms mediating the causal connection in question. A possible exception would be contexts in which shared background knowledge and rules of conversational implicature lead to a claim asserting more than what it, would seem to when taken purely at face value.

3.2.1 Empirical inspiration for an AcMe theory: causal perception and belief

In what follows, I assume a close connection between the content of our adult concept of causation, and the beliefs in which it figures, and the contents of causal claims.[4] Like Woodward, I take research into causal perception in infants and young children to be a good starting point from which to launch an investigation into the origin and character of our adult concept of causation (cf. Woodward forthcoming). Particularly important in this regard is research into how infants perceive so-called *launch events* in which one simple object in a display is seen to move off, billiard-ball style, immediately after contact with another (Leslie and Keeble 1987; Schlottmann and Shanks 1992). As a point of comparison, launch events seem, for adults, to automatically elicit vivid, cognitively impenetrable impressions of causal interactions, impressions easily dispelled by interposing a spatial or temporal gap between the end of the first object's movement and the start of second's. The way in which infants attend to launch event lends some credence to the view that infants have a similar impression of these events.

A set experiments frequently cited in support of this proposal involves habituating infants to either a launch event or an otherwise identical no-contact event—that is, one type of event is presented repeatedly until there is a substantial drop-off in the amount of attention (measured by looking time) that infants pay to it (Leslie and Keeble 1987; Scholl and Tremoulet 2000). Following habituation to one kind of event, infants are presented with the other, resulting in a relative increase in the amount of attention paid to the latter. This demonstrates, minimally, that infants can discriminate between the two sorts of stimuli. Subsequent research suggests that it is not just differences in the spatiotemporal arrangements of the items that drives dishabituation; rather there seems to be something distinctive about (apparent) causal interactions that especially garners infants' attention. Leslie and Keeble (1987) show, for instance, that a change in the direction of motion (e.g., from rightward to leftward) of the items involved produces significantly less dishabituation if the there is no contact between the items than if there is. This, they argue, may be due to the fact that there a switch in the items' roles (i.e., from agent of causation to patient and vice versa) only in the contact events. Oakes and Cohen (1990) have shown, further, that where there is a either temporal or spatial gap between the end of the first object's motion and the start of second's, habituation carries over from one of kind of non-causal sequence to the other but not from either non-causal sequence to the (apparent) causal sequence. It appears, then, that infants specifically discriminate (apparent) causal from otherwise identical non-causal sequences. However, this still does not tell us precisely what is going on in cases where an infant (or even an adult) perceives a causal interaction—that is, it does not yet tell us what the distinctive content is of causal perception.

At least with regard to the findings cited above, one might put forward a minimalist *expectation-based* interpretation according to which infants come to expect that

[4] "Concept," in this context, refers to the implicit knowledge that underlies one's linguistically appropriate deployment of a term and that in some way contributes content to one's thoughts about the term's referent. Externalism about linguistic content might initially give one pause regarding the connection between the contents of causal beliefs and causal claims, but "cause" is not a natural kind term, at least not in the potentially troublesome sense of its referent having a non-superficial essence that one might expect experts (e.g., causation scientists) to reveal.

certain types of spatiotemporal sequences will be followed by others (see Woodward forthcoming).[5] In fact, the expectation that certain types of events will be followed by others seems to be at least a part of what is involved in causal perception in infants. There is, for instance, a well-documented tendency for infants to look longer at collision events when they have an unlikely or impossible outcome (Baillargeon 1994). They appear to be in some manner or other surprised by such events and to expect a different outcome.

One might, on the other hand, give a *gestalt* interpretation of the perception of launch events according to which content is added to the available sensory evidence in much the way that it might be added in the case of other perceptual illusions—for instance, much as illusory contours seem to be added to the immediate sensory data when viewing the Kanizsa triangle. The proposal that something similar may hold with regard to the perception of launch events does gain traction from the aforementioned fact that, for adults, launch events are like other perceptual illusions in eliciting a vivid, cognitively impenetrable impression, one that can dispelled by even minor alterations of the sensory data. Taken only this far, however, the gestalt interpretation fails to tell us precisely what the distinctive content is of causal perception. One possibility is that the added content involves the projection of unseen forces or bodily sensations of pushing, pulling, and the like onto an experienced scene (see Wolff 2007, p. 86).

One might also offer a far more opulent *causal belief* interpretation according to which the distinctive of the perception of launch events in infants is due to the application of a full-blown concept of causation, a concept at least much like the one utilized by adults when they think and reason about causal sequences. As Woodward (forthcoming) points out, this interpretation of causal perception appears to be undermined by the fact that infants and some non-human primates are able to distinguish causal from non-causal sequences, and yet they fail to exhibit the sorts of behaviors one would expect, and that older children exhibit, if they were forming beliefs about the causal structure of the world. The causal belief interpretation is also called into question by the fact that causal perception and causal belief sometimes deliver conflicting verdicts about the world. For instance, superficial justificatory information (e.g., where a change in the color of one object is reliably followed by motion in another) can lead one to judge that the first happening causes the second even though it does not *look* that way. Likewise, it may look like a causal connection is present even though superficial (as Woodward terms it) "contingency" information—that is, information about whether or not each happening occurs under variations to the other—leads us to believe otherwise. We should bear in mind, however, that this double dissociation of causal perception and causal belief does not have specifically to do with cases where belief is driven by superficial contingency information. It has been observed with regard to deeper, mechanistic information in, for instance, a cross-sectional study conducted by Schlottmann (1999).

In one of Schlottmann's experiments, subjects from four different age groups (ages 5, 7, 10 and adults) saw two balls dropped into separate holes in a box, the second ball being dropped 3 s after the first. Immediately after the second ball was dropped,

[5] For our purposes, it matters not whether the expectations are innate, due to learning biases, or are purely data driven.

a bell rang from inside the box. Subjects of all ages exhibited a heavy preference for saying that the second ball caused the bell to ring, a tendency which was taken to reflect the triggering of causal perception by temporal contiguity information.[6] In the next phase, subjects were familiarized with two mechanisms, depicted in Fig. 1a and b. For reasons that should be apparent, dropping the ball into the see-saw device will make the bell ring very quickly whereas dropping it into the ramp device will make it ring after a delay. Even the youngest age group seemed to quickly understand all of this and, after placing a particular device under one of the holes in the box, they (with a few errors on the first trial among 5 year olds) were able to correctly predict how long it would take for the bell to ring. In the final phase, subjects saw that (but not where) one or the other device was put in the two-hole box. The balls were dropped as in the first phase, with a 3 s delay between them. The key finding has to do with when subjects saw the ramp device being put in the box. Adults in this condition consistently judged that whichever of the two balls was dropped first was the one that caused the bell to ring. In contrast, 5 year olds mostly judged that the second ball dropped (which was temporally contiguous with the effect) caused the bell to ring. In other words, despite understanding how the ramp device works and knowing that it was inside the box, their causal judgments were driven by temporal contiguity information rather than mechanism information. 10 year olds performed much like adults on this task, and 7 year olds exhibited an intermediate level of performance.

It appears that at a young age causal perception takes precedence over explicit mechanism information in guiding causal belief, but later in life the opposite pattern obtains.

On one reasonable interpretation of the foregoing evidence, causal perception is triggered by certain forms of temporal contiguity information (e.g., involving spatial, sonic, and presumably other properties) and the application of our concept of causation as occurs in cases of causal belief is triggered either by causal perception (preferentially so in young children) or by either superficial or deep justificatory information (preferentially so in adults). It hard to fathom how, even on the gestalt interpretation, the content of that concept might issue directly from the content of causal perception in such as way as to give rise to this complicated set of application conditions (Woodward forthcoming). On the other hand, if one backs away from the claim that the content of our concept of causation derives *directly and exclusively* from the content of causal perception—for instance, if one allows that causal perception is just one facet of a complicated bootstrapping story—one may find that even the austere expectation-based interpretation of the content of causal perception suffices (i.e., for the purpose at hand).

To see how, recall that at least a part of the distinctive content of causal perception is the expectation that a certain type of event will be followed by another. Expectations of this sort are, moreover, not restricted to paradigmatic causal interactions (e.g., collisions), as infants also have expectations about many other facets of the physical world. For instance, they seem to have the following expectations:[7]

[6] This putative instance of causal perception is, it bears noting, not triggered purely by spatiotemporal properties, but also by sonic ones.

[7] See Baillargeon (1994), Schlottmann (1999), Scholl and Tremoulet (2000), Woodward (forthcoming).

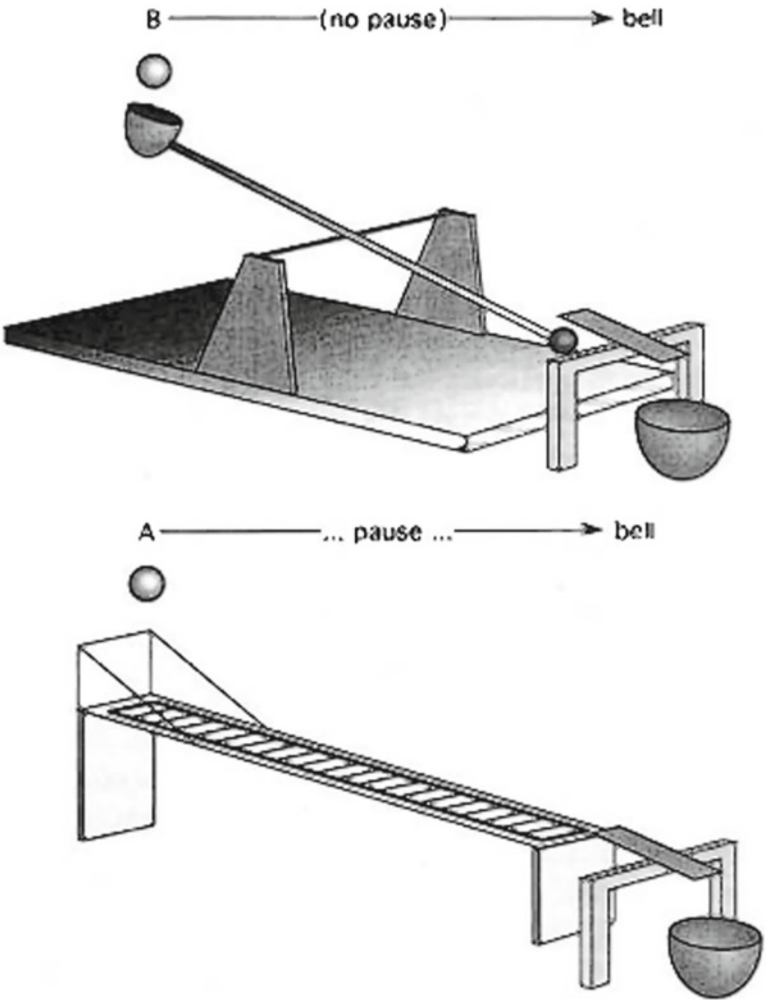

Fig. 1 Two ways of making a bell ring (from Schlottmann 1999). **a** (*top*): Ball falling into cup forces far end of lever up, knocking object off perch and into brass cup. **b** (*bottom*): Ball falls onto tracks, rolls down, and knocks object off perch and into brass cup

Objects will not cease to exist when passing behind an occluding object (persistence).
Solid objects will not pass through one another (impenetrability).
Objects without support will fall (support).
The parts of an object will move in unison (cohesion).
The interposing of a stout barrier will prevent one object from colliding with another (blocking).

These expectations become progressively refined over the course of infancy so as, for instance, to eventually include the following (Baillargeon 1994):

> When (keeping the material substratum constant) the size of an object is increased, there will be an increase in the distance to which a wheeled object that it strikes will role.
> An object will fall if the proportion of its bottom surface that hangs over its support structure is too large.
> An object of one fixed size cannot fit inside of another of a smaller fixed size.

It would, in addition, be quite surprising if infants did not also come to refine their expectations regarding how the various materials from which objects are constructed affect their behaviors and interactions. While our adult concept of causation probably does not arise solely from the content of the kind of causal perception associated with launch events, it may well arise from the integrated use of an array of perceptual, or perceptually derived, expectations regarding spatial, temporal, and other properties.

Consider, for instance, that independent research into the human ability to reason about mechanical systems suggests that we often understand how the occurrence of a particular event in a mechanical system (e.g., a series of interlocking gears) makes another happen by chaining together in thought or imagination *primitive*—which is to say that the reasoning process does not incorporate any deeper information about the mechanisms underwriting or mediating the behaviors in question—expectations regarding how one happening leads to another (Schwartz and Black 1996). Early in development, these may the very same expectations at issue in research into the development of the perception of causation, impenetrability, support, and the like (Leslie and Keeble 1987; Schlottmann 1999; Hegarty 2004). There is, in fact, a fair amount of evidence that mechanical reasoning utilizes some of the same systems that are involved in perceptual processing and, in addition, that it often has more of a depictive character than a descriptive character (Hegarty 2004). Such considerations lead many researchers to speak of mechanical reasoning as involving, at least often, the 'running' of mental simulations of events (see Waskan 2006).

Recall also that even the youngest children in the Schlottmann study seemed able, following exposure to the two devices, to understand why dropping the ball into the hole under which had been placed the device in 2A makes the bell ring quickly and why dropping it into the hole under which had been placed the device in 2B makes the bell after a delay. The research reviewed here suggests one plausible account of how it is that children come, at least initially, to understand how *one event in a particular mechanical system makes another happen*: It is the outcome of representing the (at least) spatiotemporal arrangement of the relevant objects and chaining together perceptual expectations regarding their behaviors. When this form of understanding arises as events are unfolding and the machinations are visible, one might say it involves 'on-line' integration of (at least) spatiotemporal information and various perceptual expectations. When this understanding takes place after the fact or when the known machinations have been hidden, one might say it involves 'off-line' integration (a simulation) of the same information.

While representations of these sorts would have as their content the actual sequence of events, they might in addition enable children to infer, as needed, whether or not any of countless further circumstances—for instance, placing a stout barrier (blocking) squarely on the ramp (support) of the device in 2B—would interfere with the envisioned process. One might say that the simulation embodies *explicit* knowledge of the actual sequence but also tremendous amounts of *tacit* knowledge—that is, knowledge that is not encoded explicitly but can be generated on demand—of the conditions that might interfere with that process (Waskan 2008). In the same way, an off-line simulation might also embody tacit knowledge of the fact that (provided there is not some other event that would also make the bell ring) the ringing will not occur if the dropping of the ball is prevented or altered in various possible ways (e.g., its trajectory is diverted away from the ramp, it is thrown with great force into the hole, etc.). Thus, a child would also be able to fallibly infer that the device in 2B is under one hole in the box not just by what happens when the ball is dropped into that hole but by what happens when the ball is not dropped, etc.—that is, on the basis of various sorts of contingency information.

This proposal strikes me as plausible and consistent with at least a great deal of the relevant empirical literature, and it is not unprecedented. If this were, indeed, the basis for our early understanding of how specific events in specific mechanical systems make other events happen, it is not a big stretch to suppose that we might, over the course of many such episodes, also form some notion that there is a more general category of circumstances under which one event makes another happen. Let us call the concept in question that of one happening (X) *mechanically necessitating* (MeNe-ing) another (Y). It is, unfortunately, not patently obvious what the precise content of this concept would be, but early in childhood it might amount to the idea that there is a class of circumstances in which the prior occurrence of X in some underlying or mediating system of (at least) spatiotemporally arranged parts is connected to the subsequent occurrence of Y through a chain of expected behaviors of the aforementioned sort (e.g., regarding impenetrability, collisions, support, etc.). The "absent impediment" qualifier simply signals the recognition, exemplified above with regard to the device in 2B, that the MeNe-ing relation in question can be defeated in various ways; it is the counterpart of "Ap" in "ApCoInD." It is also in keeping with this view to suppose that children might, also based upon their experiences with specific mechanisms, take various sorts of contingency information—for instance, that (barring other MeNers) Y will not occur if X is prevented and that (barring bi-directional MeNe-ing such as occurs with interlocking gears) preventing Y will not alter X—to be implied by X MeNe-ing Y. Thus children might fallibly confirm or refute their beliefs, suspicions, etc. by observing the results of changes (perhaps of their own making) to X and Y. In keeping with the above, we might say that the mental states in question explicitly (albeit in highly generic form) represent the actual relationship between X and Y, but they also have implications regarding (or tacitly represent) the consequences of counterfactual interventions on X and Y. A more direct way to fallibly confirm or refute their beliefs, suspicions, etc. would, of course, be to look 'under the hood,' so to speak, and gather deep evidentiary information. Anyone who has observed children older than about 3 years of age knows that both sorts of strategies are routinely employed.[8]

The sorts of perceptual expectations formed and refined over the course of early childhood would suffice to bootstrap understanding, with regard to at least some

[8] Regarding the latter, see Bullock (1984).

mechanical systems, of how one occurrence MeNes another and, subsequently, a concept MeNe-ing. If this account is correct, one imagines that throughout our lives the set of expectations that we come to draw upon in order to understand how events in specific systems unfold, and thus the sorts that might underlie MeNe-ing more generally, will expand well beyond the basic perceptual repertoire. For instance, if, as has been suggested, superficial contingency and deep mechanical information can both clue us in to new forms of MeNe-ing, then any form so detected might itself become part of our repertoire for representing how events in any number of further systems can be expected to unfold, and so on. One would, moreover, expect that our linguistic community would, whether by helping us to master (inter alia) a great many action verbs (see Machamer 2004) or through trusted testimony, itself be a rich source of relevant expectations.

In any case, what I have been building towards is the suggestion that our concept of causation might just be the concept of MeNe-ing and that to believe, suspect, etc. that X causes Y is to believe, suspect, etc. that X MeNes Y.[9] This proposal strikes me as perfectly sensible, and even plausible, nor is it without precedent in the psychological literature (e.g., see Schlottmann 1999, Schlottmann et al. 2002). Most who have taken an interest in this literature are, for instance, familiar with Ahn and Kalish's claim that "what is essential in the commonsense conception of causal relations is the belief that there is some process or mechanism mediating this relation, *whether understood in detail or not*" (2000, p. 202 *emph. mine*). It should be stressed, however, that the MeNe theory differs from some actualist-mechanist theories of our concept of causation—for instance, Wolff (2007) and even to some extent Ahn and Kalish (2000, p. 201)—in that it rejects outright the proposal that the transfer of energy, physical force, momentum, etc. lies at the heart of our concept of causation.

It is, unfortunately, not possible here to take on the many empirical and philosophical challenges that must be dealt with by a theory of our concept of causation. Still, I would be remiss if I did not at least discuss the concerns that Woodward himself has voiced regarding the MeNe theory.[10] Some of these—for instance, the concern that causal perception does not suffice to bootstrap the adult concept of causation, that actualist-mechanistic theories do not account for the sensitivity of causal belief to contingency information, and that actualist-mechanist theories have trouble explaining high-level (e.g., biological) causation due to their reliance on low-level notions like energy or force transmission—have, I hope, been either directly or indirectly addressed by the foregoing remarks. There are, however, still two major outstanding worries.

One stems from a case, attributed to Hitchcock, in which chalk on a cue stick gets transmitted to a cue ball, and from there to an eight ball, which subsequently sinks. Woodward's ApCoInD theory seems easily to account for why it is that we judge that

[9] It might be objected that to believe that smoking causes cancer is not to believe that the occurrence of smoking always leads (even absent impediment) to the occurrence of cancer. In response one might follow Woodward's lead and tack on something like, "or at the very least an increase in the probability of Y." I have reservations regarding the force of this objection (and hence regarding the need for such an amendment) but I will not contest it here.

[10] Woodward voiced these concerns while commenting on a condensed, nascent version of this essay at the *2010 Pacific Division Meeting of the American Philosophical Association*.

the movement of the cue stick, rather than that of the chalk, is what caused the eight ball to sink: the ball sinking bears an ApCoInD relation to the movement of the stick but not to the movement of the chalk. Actualist-mechanist theories that put all of the weight on spatiotemporally continuous transmission of force or energy, on the other hand, seem to have trouble explaining why we judge that one happening is causally relevant and the other is not, for there is a spatiotemporally continuous transmission from each to the effect. Woodward worries that the MeNe theory might run aground of a similar problem. The problem, as Woodward puts it, is to explain why we take one of these mechanical relationships to be of the wrong kind.

To see how a proponent of the MeNe theory might respond, notice first that what we are dealing with here is a causal relationship of which we have some amount of deep evidentiary information. Recall, moreover, that children's expectations regarding how events unfold in specific mechanical systems become more sophisticated over time. Arguably, they come to expect that a rigid ball on a flat surface will (absent impediment) roll when struck squarely by something like a cue stick, that the velocity of the ball will depend on how fast and how heavy the stick is and on whether it hits squarely, and so on. By the same token, children surely do not expect that an impact with fast moving chalk dust will result in a cue or eight ball moving off quickly. These sorts of expectations arguably have much to do with why even children would, presumably, judge that the movement of the stick is causally relevant but that of the chalk is not. If they see or, based upon a description, are asked to simulate the sequence in question, they will understand that the cue stick moving MeNes the ball sinking through a chain of fairly basic perceptual expectations and that the chalk dust moving does not. The former, in other words, is connected to the effect by the right kind of mechanical relation.

The second major worry is that some causal relations (e.g., in biology) involve so-called "double prevention" (a.k.a. "disinhibition," "double prevention," or "negative control") (Woodward 2002, 2004). In particular, there are cases where we believe that some X causes some Z even though we know that it does so by preventing Y from preventing Z. For instance, one who accepts Mink's model (discussed earlier) might think that excitation of striatum causes (or is part of the total cause of) movement execution in that neurons of the striatum prevent neurons of SN from preventing particular motor programs from being executed. Woodward claims that counterfactual theories seem to do a better job than typical actualist-mechanist theories of capturing the sense in which such double-prevention relationships are believed to be causal and that the MeNe theory may be vulnerable to a similar worry.

It might be fairer, however, to suggest, with regard to such cases, that whereas traditional actualist-mechanist theories appear too restrictive, counterfactual theories appear too permissive. There are, after all, cases of double-prevention which would (at least prima facie) be classified as causal on the ApCoInD approach but which few people would so classify. Consider, for instance, a case where one billiard ball alters the trajectory of a second, thereby preventing it from altering the trajectory of a third, the outcome being that the third sinks into a pocket. The ball sinking in this case bears an ApCoInD relation to the motion of the first ball, but there is at least some evidence that few would take the former to have caused the latter (see Walsh and Sloman 2005; Woodward forthcoming).

The MeNe theory offers at least some promise of charting a middle course. It does not rely upon notions like *force* or *energy transfer*, though, given the sorts of behavioral expectations hypothesized to bootstrap the concept of MeNe-ing, perhaps there is a tacit restriction of contact between each link in the chain of happenings connecting cause to effect. Consider, for instance, a case where a vase is placed squarely onto a supporting surface, after which a ball is thrown and knocks the support out from under (i.e., prevents it from preventing the downward motion of) the vase, which subsequently breaks. This too is ultimately an empirical matter, but probably most would judge that throwing the ball caused the vase to break. If, however, the chain of contact is broken—for instance, if the ball knocks the same support out of the way just before the vase is released onto it—I suspect that few would judge that the ball caused the vase to break (though the thrower may still be held responsible). Notice also that in both cases the vase breaking bears an ApCoInD relation to the ball being thrown, but only in the former case is the ball being thrown connected to the vase breaking by a continuous chain of contact (adhering to fairly basic perceptual expectations) between the relevant items. Only in the former case does the ball being thrown MeNe the vase breaking.

This bodes well for the MeNe theory, but there is, unfortunately, at least one further wrinkle. Consider the following explanation for the link between smoking and cancer.

> SMOG: Certain forms of genetic damage, which are a common occurrence whether one smokes or not, initiate (if not repaired) a certain form of runaway tissue growth (i.e., malignant tumors). Cigarette smoke contains chemicals that, when inhaled, sometimes disable the cellular devices that repair the aforementioned forms of genetic damage.

It *may* be that most people would, even if they believe that the described break in the chain of contact obtains, still judge that smoking causes cancer. Insofar as the MeNe theory includes stricture that there must be a continuous chain of contact linking causes with effects, such a result would appear troubling on its face. Even so, a MeNe theorist might try to explain the result away through, for instance, one of the following proposals:

(A) Normative associations or prior beliefs about the case might counteract our natural tendency to deny a causal connection upon learning that there is a break in the chain of contact.

(B) The complexity of the SMOG model might lead us to gloss over what we would otherwise regard as a salient spatiotemporal gap between repair mechanisms and damaged genes.

(C) The action verbs (e.g., “inhale”, “repair,” “disable”) used to describe the case may set off our (fast and automatic) System 1 causation detector (see Evans 2003). On closer examination, perhaps our (more careful and deliberative) System 2 would reach a different result—namely, that normal wear and tear causes cancer and smoking just ensures that this process goes uninterrupted.

In any case, as far as double-prevention cases are concerned, it appears that neither MeNe theory nor ApCoInD theory can skate to an easy and obvious victory.

Let me also emphasize, before turning finally to the contents of causal claims, that my primary goal in offering up the MeNe theory is not to demonstrate that it is far and away the best account of our concept of causation (though I do find it quite promising). Instead, I just hope, first, to combat the feeling that the AcMe approach it exemplifies is nonsensical or hopelessly ill-defined and, secondly, to give a concrete illustration of how a worked-out AcMe theory might help to address Woodward's biggest concerns about actualist-mechanist theories of causal beliefs and, as shown presently, causal claims.

3.2.2 An unexplored egress

Under the assumption that our concept of causation captures the standard application conditions for 'cause' (and cognates) let us return now to the question of what causal claims assert. If you will recall, one of Woodward's main objections to the idea that causal claims make assertions about actual productive mechanisms was that sometimes—in particular, under purely shallow evidentiary conditions—there seems to be nothing for the putative mechanical assertions to assert. As we saw, Woodward here overlooks the possibility of (*inter alia*) a consistently non-specific (AcMe) theory of causal claims according to which "X causes Y" (typically) just asserts that X bears some general sort of mechanical relation Y. One possible way of putting flesh on the bare bones of this proposal would be to put forward a MeNe theory of causal claims according to which claims of the form "X causes Y" simply assert that one happening MeNes another. On this view, causal claims do not (typically) assert anything about specific mechanisms by which this occurs, and Woodward's concern about shallow evidentiary conditions can thus be diffused.

By highlighting how Woodward has, thanks admittedly to neglect of this alternative on the part of mechanists themselves, overlooked the possibility that an AcMe theory of causal claims is correct, I hope to at least plant seeds of doubt regarding his above criticisms of mechanistic theories *of explanation*. The MeNe theory, for example, shows how mechanists might answer concern (ii), which is that some explanations (viz., bare and contentful causal claims) are not constituted by claims about mechanisms. On any AcMe theory, claims of this sort *are* about mechanisms. On the MeNe theory, for instance, they essentially just assert that same the kind of mechanical-production relation discerned in other cases (i.e., through on-line and off-line integration of specific spatiotemporal expectations) also obtains, if only covertly, in the case at hand. Claims of this sort do not make assertions about specific mechanisms, though, and so they convey at best only the wispiest understanding of the target happening. Thus, to the extent that they do pass muster as explanations, it is only just barely.

As for (i), we saw that on Woodward's view a major impetus for taking mechanism explanations to be constituted by networks of assertions about counterfactuals is that they bottom out at claims about how the parts of the system being described causally interact. However, insofar as mechanists are able to address (ii), they can also address (i). A proponent of the MeNe theory of causal claims might, for instance, respond that mechanism explanations do bottom out at causal claims but that, instead of making assertions about counterfactuals, these claims make further, non-specific assertions about mechanisms. Mechanism explanations bottom out at claims asserting that, but not how, on happening MeNes another. At the same time, one need not, on this view, deny that information about counterfactuals is any part of the contents of bottom-out claims. On the MeNe theory, causal claims explicitly assert that one happening MeNe's another, but, as explained in the previous section, they may also have implications regarding, or *tacitly* assert, what will happen under counterfactual circumstances.

4 Conclusion

As we also saw, (i) formed the basis for Woodward's claim that mechanism explanations reduce to networks of claims asserting ApCoInD relations, and (ii) formed the basis for his claim that mechanistic theories should be relegated to handling only a proper subset of explanations. Whether on not mechanists about explanation embrace the MeNe theory of either the concept of causation or of the contents of causal claims, my hope is that the foregoing considerations will at least convince them that pursuit of a non-specific, actualist-mechanist theory of causal claims (of whatever sort) offers a way, one which has yet to be fully explored, in which they might stave off Woodward's apparent attempt at marginalization and assimilation.

Acknowledgements Thanks to Jim Woodward, Gaultiero Piccinini, Brendan Shea, Daniel Estrada, and anonymous referees at *Synthese* for their helpful comments.

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